
Algorithm 1 Overall Risk-aware Decision Procedure

Require: Scenario x_t , L0 experience store $\mathcal{M}^{L0}$, L2 rule store $\mathcal{M}^{L2}$

Ensure: Approved action a_t^{app} and decision trace $\mathcal{T}_t$

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1: for  $k \in \{\text{Advocate, Critic, Decision}\}$  do
2:    $\mathcal{B}_k \leftarrow \text{RETRIEVERISKMEMORY}(x_t, \mathcal{M}^{L0}, k)$ 
3: end for
4:  $\mathcal{Q}_t \leftarrow \text{MATCHRISKRULES}(x_t, \mathcal{M}^{L2})$ 
5:  $c_t \leftarrow \text{RESOLVERULECONFLICTS}(\mathcal{Q}_t)$ 
6: if  $c_t = \text{conflict}$  then
7:    $\hat{a}_t^{\text{LLM}} \leftarrow \text{safe\_stop}$ 
8:    $a_t^* \leftarrow \text{safe\_stop}$ 
9: else
10:   $d_t^{\text{adv}} \leftarrow \text{TASKADVOCATE}(x_t, \mathcal{B}_{\text{adv}}, \mathcal{Q}_t)$ 
11:   $d_t^{\text{cri}} \leftarrow \text{RISKCRITIC}(x_t, \mathcal{B}_{\text{cri}}, \mathcal{Q}_t)$ 
12:   $d_t^{\text{dec}} \leftarrow \text{SAFETYDECISION}(x_t, \mathcal{B}_{\text{dec}}, \mathcal{Q}_t, d_t^{\text{adv}}, d_t^{\text{cri}})$ 
13:   $\hat{a}_t^{\text{LLM}} \leftarrow d_t^{\text{dec}}.\text{selected\_action}$ 
14:   $\mathcal{A}_t^{\text{safe}} \leftarrow \text{FILTERUNSAFEACTIONS}(x_t, \mathcal{Q}_t)$ 
15:  for all  $a \in \mathcal{A}_t^{\text{safe}}$  do
16:     $u_a \leftarrow \text{TASKUTILITY}(a)$ 
17:     $r_a \leftarrow \text{ACTIONRISK}(x_t, a, \mathcal{B}_{\text{dec}})$ 
18:     $q_a \leftarrow \text{ACTIONUNCERTAINTY}(x_t, a, \mathcal{B}_{\text{dec}}, d_t^{\text{adv}}, d_t^{\text{cri}}, d_t^{\text{dec}})$ 
19:  end for
20:   $\mathcal{E}_t \leftarrow \{a \in \mathcal{A}_t^{\text{safe}} \mid r_a \leq \tau_R \wedge q_a \leq \tau_Q\}$ 
21:  if  $\mathcal{E}_t \neq \emptyset$  then
22:     $a_t^* \leftarrow \arg \max_{a \in \mathcal{E}_t} u_a$ 
23:  else
24:     $a_t^* \leftarrow \text{safe\_stop}$ 
25:  end if
26: end if
27:  $a_t^{\text{app}} \leftarrow \text{SAFETYGUARD}(x_t, a_t^*)$ 
28:  $\mathcal{T}_t \leftarrow \text{BUILDDECISIONTRACE}(\mathcal{B}, \mathcal{Q}_t, \hat{a}_t^{\text{LLM}}, a_t^*, a_t^{\text{app}})$ 
29: return  $a_t^{\text{app}}, \mathcal{T}_t$ 
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